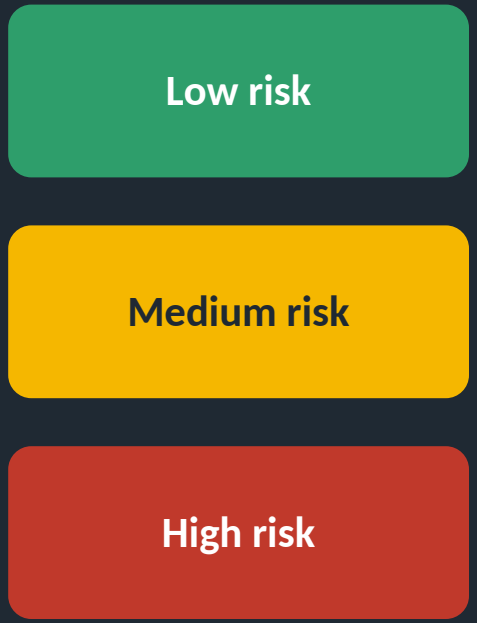
MACHINE LEARNING · B.TECH CSE · A.Y. 2026-27

Road Segment Risk Classification Using Decision Trees and LightGBM

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Classifying road segments into risk categories

Why road risk needs prediction

1

Traffic volume

How heavily a segment is used

2

Road characteristics

Layout and physical condition

3

Weather conditions

Environmental factors on the road

4

Accident history

Past accident-related information

OBJECTIVE

Classify road segments into risk categories and find the most effective model for accurate risk prediction.

Helps authorities improve road safety and reduce accidents.

Data and preprocessing

Road segment dataset

- Traffic volume
- Road characteristics
- Weather conditions
- Accident-related information

1. **Handle missing values**
2. **Encode categorical attributes**
3. **Scale the required features**
4. **Split into training and testing sets**

Two models, one comparison

Decision Tree

Single-tree classifier

- Splits data with if-then rules
- Easy to read and interpret
- Learns one tree from the training data

LightGBM

Light Gradient Boosting Machine

- Ensemble of boosted decision trees
- Combines many learners for stronger predictions
- Handles complex feature interactions

Both models are trained on the same training set and tested on the same testing set

Evaluation and results

Accuracy

Correct predictions overall

Precision

Correctness of predicted risks

Recall

Share of true risks found

F1-score

Balance of precision and recall

Both models work

Decision Tree and LightGBM both classify road segment risk effectively.

LightGBM performs better

Higher accuracy and better overall performance, thanks to its ensemble learning capability.

Conclusion

- Machine learning can classify road segment risk effectively
- LightGBM outperforms the Decision Tree
- Supports proactive road safety planning and preventive measures by transport authorities

SOFTWARE STACK

Python 3.x	Jupyter / Colab
Pandas	NumPy
Scikit-learn	LightGBM
Matplotlib	Seaborn
Windows	